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END SEMESTER EXAMINATION, DECEMBER 2025

Name of the Course: B.Tech Biotechnology

Semester: III

Name of the Paper: Sustainable Agriculture and Organic Farming for Food Security

Paper Code: TBT 306c

Time: 180 minutes

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question.
- (iii) Each question carries 10 marks.

Q1.	(20 Marks)	
(a)	Discuss role of biotechnology to help develop robust sustainable agroecosystems?	CO3
(b)	Discuss different techniques utilized for managing the soil health? Shed a light on how conservation practices can be useful for land use management	
(c)	Discuss how is rotation design of crops utilized to boost productivity in organic agriculture?	
Q2.	(20 Marks)	
(a)	Discuss about (i) compost production and (ii) vermicomposting	CO1
(b)	What is sustainable agriculture? Discuss how can we shift towards it?	
(c)	What is the global food security challenge and how it is aligned with UNSDG2 goal?	
Q3.	(20 Marks)	
(a)	Discuss about international trade regulations and how India regulates its food standards?	CO4
(b)	Discuss USDA/EU regulations and how are these organic certified products made available in the market	
(c)	Write about Food Safety Standards Authority of India (FSSAI) and its roles	
Q4.	(20 Marks)	

(a)	How can organic food products be promoted?	CO2
(b)	Discuss any two important methods of organic crop management ?	
(c)	What do you understand by precision agriculture? Elucidate digital technologies that are being used for organic food production	
Q5.	(20 Marks)	
(a)	What is the public perception of organic food?	CO5
(b)	Discuss NPOP (National Program for Organic Production)?	
(c)	Analyze a real-world example of a country or region that has successfully increased organic production.	